

Chemistry

CYL1010



Instructor: Dr. Suresh Sarkar

Email: sarkar@iitj.ac.in

Office: Chemistry Department, #319

Contents

Title	Chemistry	Number	CYL1010
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. 1 st Year	Type	Compulsory
Prerequisite	Nil		
Objectives 1. Relate the fundamental principles of chemistry with practical problems encountered for engineers and place the emphasis on problem-solving 2. Provide knowledge of engineering materials metals, composites, polymers, cement, steel, fuels, batteries and drugs. Learning Outcomes 1. Understanding the behavior of matter and materials using fundamental knowledge of their nature. 2. Understanding the basic concepts and implementation of the same in advanced and interdisciplinary areas. 3. Predict potential complications from combining various chemicals or metals in an engineering setting. Contents <i>Atomic and Molecular Structure [6 Lectures]:</i> particle in a box, angular momentum, wavefunction, multielectron atoms, homonuclear and heteronuclear diatomic molecules, non-covalent interactions <i>Solid State Chemistry [6 Lectures]:</i> Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties. <i>Surface Chemistry[3 Lectures]:</i> Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension. <i>Electrochemical Systems [5 Lectures]:</i> Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion <i>Basic Spectroscopy [7 Lectures]:</i> Selection Rules, Fundamentals FTIR, Raman, UV-visible and Fluorescence spectroscopy <i>Transition Metal Chemistry [5 Lectures]:</i> Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry <i>Organic Chemistry [8 Lectures]:</i> Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies. Textbooks 1. Silberberg, M., <i>Chemistry: The Molecular Nature of Matter and Change</i>, 6th Edition, McGraw Hill Education. 2. West, A.R., (2015), <i>Solid State Chemistry and Its Applications</i>, 2nd edition, John Wiley & Sons. 3. J. D. Lee, <i>Concise Inorganic Chemistry</i>, 5th Edition. 4. Atkins, <i>Physical Chemistry</i>, 11th Edition. Reference Books 1. McMurry, J. E. & Fay, R. C. <i>Chemistry</i>, 5th Edition, Pearson. 2. Jonathan Clayden, Nick Greeves, Stuart Warren, <i>Organic Chemistry</i>, 2nd Edition. 3. F. Albert Cotton, Geoffrey Wilkinson, <i>Basic Inorganic Chemistry</i>, Wiley Student Edition. Self Learning Material 1. https://nptel.ac.in/downloads/122101001/			

Contents

- **Surface Chemistry[3 Lectures]:** Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension.
- **Solid State Chemistry [6 Lectures]:** Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties.
- **Electrochemical Systems [5 Lectures]:** Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion



Relation of E° and ΔG°

E° is related to ΔG° , the free energy change for the reaction.

$$\Delta G^\circ = -n F E^\circ$$

- F = Faraday constant = $9.6485 \times 10^4 \text{ J/V}\cdot\text{mol}$
- n = the number of moles of electrons transferred.



Michael Faraday
1791-1867

Discoverer of

- electrolysis
- magnetic props. of matter
- electromagnetic induction
- benzene and other organic chemicals



n for Zn/Cu cell ?

$$n = 2$$

Nernst Equation

□ Have relationship between Gibbs Free Energy and Equilibrium constant:

$$\Delta G^\circ = -RT \ln K$$

□ ΔG for a reaction depends on the concentration by:

$$\Delta G = \Delta G^\circ + RT \ln Q \quad \text{where } Q \text{ is the reaction quotient} = \frac{[\text{product}]}{[\text{reactant}]}$$

$$\Delta G = -n F E_{\text{cell}} \quad \text{and} \quad \Delta G^\circ = -n F E_{\text{cell}}^\circ$$

$$-n F E_{\text{cell}} = -n F E_{\text{cell}}^\circ + RT \ln Q$$

Dividing across by nF gives:

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{nF} \ln Q$$

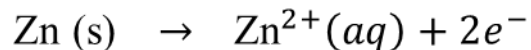
Nernst Equation

i.e. the cell potential at any conditions depends on the potential under standard state conditions and a term for the potential at nonstandard-state conditions

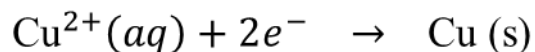
How do we write the Nernst equation for Daniell cell?

Half-Reactions

- Anode (oxidation):**



- Cathode (reduction):**



Standard EMF

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

From standard reduction potentials:

- $E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} = +0.34 \text{ V}$
- $E_{\text{Zn}^{2+}/\text{Zn}}^{\circ} = -0.76 \text{ V}$

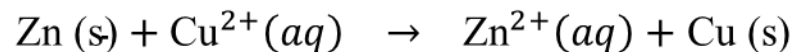
So:

$$E_{\text{cell}}^{\circ} = 0.34 - (-0.76) = 1.10 \text{ V}$$

Classical Daniell cell:



Overall Reaction



- Electrons exchanged (n): 2**

Reaction Quotient (Q)

$$Q = \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$$

**Solid Zn and Cu activity = 1,
So, they drop out**

Nernst Equation for the Cell

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log Q$$

Substitute $n = 2$:

$$E_{\text{cell}} = 1.10 - \frac{0.0591}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$$

- If $[\text{Zn}^{2+}]$ increases $\rightarrow Q$ increases $\rightarrow$ EMF decreases.
- If $[\text{Cu}^{2+}]$ increases $\rightarrow Q$ decreases $\rightarrow$ EMF increases.

Relationships between ΔG° , K , E°_{cell}

Spontaneity of Redox Reactions

Table 18.2 Relationships Among ΔG° , K , and E°_{cell}

ΔG°	K	E°_{cell}	Reaction Under Standard-State Conditions
Negative	>1	Positive	Favors formation of products.
0	$=1$	0	Reactants and products are equally favored.
Positive	<1	Negative	Favors formation of reactants.

$$\Delta G^0 = -RT \ln K = -nFE_{\text{cell}}^0$$

Application of EMF measurement

1. Determination of equilibrium constant
- 2 Determination of solubility of sparingly soluble salt
3. Determination of Valence
4. Determination of thermodynamic functions
5. Determination of pH of a solution
6. Potentiometric titrations
7. Determination of activity coefficient
8. Determination of transference number

How? Detail description with picture as per required

1. Determination of Equilibrium Constant

Electromotive force (EMF) measurements are commonly utilized in various scientific fields, including chemistry, to determine the equilibrium constant of a chemical reaction. The equilibrium constant (K) is a fundamental parameter that quantifies the extent of a chemical reaction at equilibrium



$$K = \frac{\overbrace{[C]^c [D]^d}^{\text{products}}}{\underbrace{[A]^a [B]^b}_{\text{reactants}}}$$

$$E = E^\circ - (RT / nF) \ln(Q)$$

Where:

- **E** = Cell potential (EMF) at non-standard conditions (V)
- **E°** = Standard cell potential (V) - This is a constant value specific to the reaction under standard conditions (1 atm pressure, 25°C, and 1 M concentration for all participating species).
- **R** = Gas constant (8.314 J/mol*K)
- **T** = Temperature (Kelvin)
- **n** = Number of electrons transferred in the reaction
- **F** = Faraday's constant (96485 C/mol)
- **Q** = Reaction quotient - This is the ratio of the product concentrations raised to their stoichiometric coefficients, divided by the reactant concentrations raised to their stoichiometric coefficients (unitless).

At equilibrium the cell potential is zero and $Q = K$. Substituting these into the equation we get, $0 = E^\circ_{\text{cell}} - (0.0257/n) \ln K$

$$E^\circ_{\text{cell}} = \frac{0.0592}{n} \log K$$

E° is known, then you calculate K

At **equilibrium**, the forward and reverse reactions occur at **equal rates**. That means:

- **No net reaction** is occurring anymore.
- **No net transfer of electrons** is happening.

Determination of Solubility of Sparingly Soluble Salt

Sparingly soluble salts (like AgCl , BaSO_4) have very low solubility in water. The **EMF method** uses an **electrochemical cell** to measure the **ion concentration** in equilibrium with the solid salt, from which solubility can be calculated.

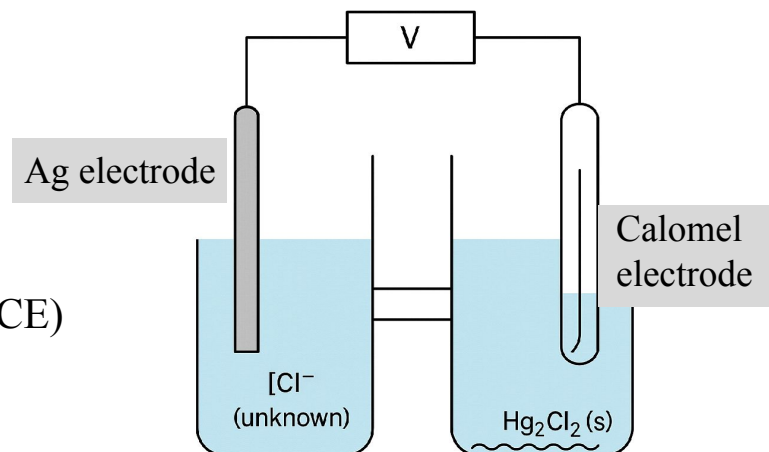
The principle relies on the **Nernst equation**, which relates the electrode potential to the ion activity (or concentration) in solution.

Take AgCl as an example:



Electrochemical Cell Setup

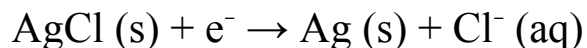
- $\text{Ag} \mid \text{AgCl (s)}, \text{Cl}^- (\text{unknown}) \parallel \text{Reference electrode (SCE)}$
- **Indicator electrode:** Ag/AgCl responds to $[\text{Cl}^-]$.
- **Reference electrode:** fixed potential.



- **Left electrode:** Silver electrode in contact with solid AgCl and solution containing Cl^- .
- **Right electrode:** Standard reference electrode (like saturated calomel electrode, SCE).

Cell Reaction

The half-cell reaction at the silver electrode:

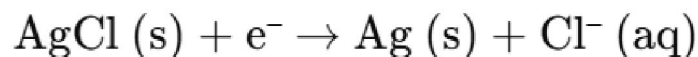


- The electrode is reversible with respect to **Cl^- ions**.
- That means the electrode potential depends directly on the **chloride ion activity (concentration)** in the solution.

The EMF of the cell depends on **chloride ion concentration**:

Standard electrode potential

The standard reduction potential for the half-reaction:



is defined as $E_{\text{Ag/AgCl}}^0$.

The **general Nernst equation** is:

$$E = E^0 - \frac{0.0591}{n} \log Q$$

where

- E = electrode potential under given conditions
- n = number of electrons transferred (here $n = 1$)
- Q = reaction quotient

(Solid Ag and solid AgCl have activity = 1, so they drop out).

For our half reaction:

$$Q = \frac{a_{\text{Cl}^-}}{1}$$

Thus:
$$E = E_{\text{Ag/AgCl}}^0 - 0.0591 \log [\text{Cl}^-]$$

$$[\text{Cl}^-] = 10^{(E - E_{\text{Ag/AgCl}}^0)/0.0591}$$

- **If $[\text{Cl}^-]$ increases** → The logarithmic term increases → The potential becomes **more negative**.
- **If $[\text{Cl}^-]$ decreases** (like in a saturated solution of AgCl) → The potential becomes **more positive**.
Therefore, **the electrode potential (and hence the cell EMF) depends only on the chloride ion concentration**.

Since $AgCl \rightleftharpoons Ag^+ + Cl^-$ and in pure water:

$$[Ag^+] = [Cl^-] = s$$

where s is the **solubility of AgCl**. Thus, the **solubility S** is directly obtained from the EMF measurement.

Example

- Reference electrode = SCE ($E^\circ = +0.241$ V vs SHE).
- Measured cell EMF = 0.200 V.
- $E^\circ_{Ag/AgCl} = +0.222$ V.

Then:

$$E_{\text{cell}} = E_{Ag/AgCl} - E_{SCE}$$

$$0.200 = (0.222 - 0.0591 \log[Cl^-]) - 0.241$$

$$0.200 = -0.019 - 0.0591 \log[Cl^-]$$

$$0.219 = -0.0591 \log[Cl^-]$$

$$\log[Cl^-] = -3.71$$

$$[Cl^-] = 1.95 \times 10^{-4} \text{ M}$$

So the chloride concentration is $\sim 2 \times 10^{-4}$ M.

Why it doesn't work for highly soluble salts (like NaCl, KNO₃, etc.)?

(a) Very high ion concentration

- Highly soluble salts completely dissociate.
- Ion concentrations are **large (0.1 M to several M)**.
- Nernst equation uses **activity**, not raw molar concentration:

$$E = E^0 - \frac{0.0591}{n} \log a_{\text{ion}}$$

At high concentration, $a \neq [C]$

- Accurate EMF-to-concentration conversion becomes unreliable.
- Activity coefficients deviate strongly from 1.

(b) Non-ideal behavior

- In concentrated solutions, ions strongly interact → ion pairing, shielding, hydration effects.
- Debye–Hückel theory corrections become complex.
- Thus EMF cannot be straightforwardly related to solubility.

(c) No real "solubility limit" to measure

- Sparingly soluble salts reach equilibrium with a solid phase → fixed, definite $[C][C][C]$.
- Highly soluble salts (like NaCl) dissolve so much that practical “saturation” is hard to measure via EMF; precipitation only happens at very high concentrations (near saturation).
- EMF method is sensitive to small ionic concentrations but **not reliable in highly concentrated media**

Determination of Valence

Determining the valence of an element or ion is a crucial aspect of understanding its chemical behaviour and properties. Electromotive force (EMF) measurements can be applied to determine the valence of an element or ion, especially in electrochemical cells where redox reactions occur.

The **general Nernst equation** is:

$$E = E^0 - \frac{0.0591}{n} \log Q$$

where

- E = electrode potential under given conditions
- n = number of electrons transferred (here $n = 1$)
- Q = reaction quotient

For any reaction, if you know E^0 and Q , you know n

Application of EMF measurement

1. Determination of equilibrium constant
- 2 Determination of solubility of sparingly soluble salt
3. Determination of Valence
4. Determination of thermodynamic functions
5. Determination of pH of a solution
6. Potentiometric titrations
7. Determination of activity coefficient
8. Determination of transference number

How? Detail description with picture as per required

1. Relation between EMF and Gibbs Free Energy

For a cell reaction involving n moles of electrons:

$$\Delta G = -nFE$$

- n = number of electrons transferred
- F = Faraday constant ($96,485 \text{ C}\cdot\text{mol}^{-1}$)
- E = EMF of the cell (measured experimentally)

So, from the measured EMF, ΔG can be obtained directly.

2. Standard Gibbs Free Energy Change

If EMF is measured under **standard conditions** (1 M, 1 atm, 298 K):

$$\Delta G^\circ = -nFE^\circ$$

3. Relation to Equilibrium Constant

At equilibrium:

$$\Delta G^\circ = -RT \ln K$$

So combining:

$$E^\circ = \frac{0.0591}{n} \log K$$

Thermodynamics functions

4. Determination of Enthalpy (ΔH) and Entropy (ΔS)

We use temperature dependence of EMF.

(a) Gibbs–Helmholtz equation

$$\Delta G = \Delta H - T\Delta S$$

Since $\Delta G = -nFE$:

$$-nFE = \Delta H - T\Delta S$$

(b) Entropy change (ΔS)

Differentiate EMF with respect to T:

$$\left(\frac{\partial E}{\partial T}\right)_P = \frac{\Delta S}{nF}$$

So:

$$\Delta S = nF \left(\frac{\partial E}{\partial T}\right)_P$$

(c) Enthalpy change (ΔH)

From Gibbs equation:

$$\Delta H = \Delta G + T\Delta S$$

Substitute:

$$\Delta H = -nFE + T \cdot nF \left(\frac{\partial E}{\partial T}\right)_P$$

Determination of Thermodynamic Functions from EMF

Function	Relation from EMF
ΔG	$-nFE$
ΔG°	$-nFE^\circ$
K	$E^\circ = \frac{0.0591}{n} \log K$
ΔS	$nF \left(\frac{\partial E}{\partial T} \right)_P$
ΔH	$-nFE + TnF \left(\frac{\partial E}{\partial T} \right)_P$

Determination of pH of a solution

Principle

pH is defined as: $\text{pH} = -\log[H^+]$

The EMF then directly relates to the pH through the **Nernst equation**.

(b) Glass Electrode Method (most practical in labs)

Cell:

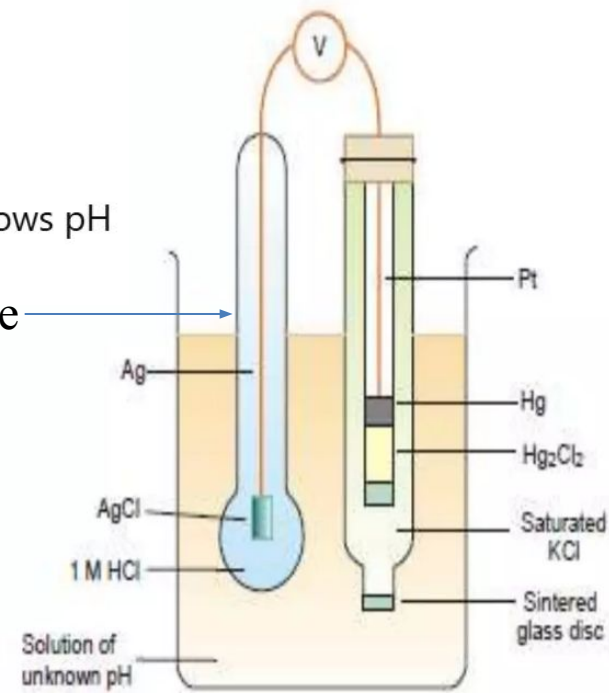
Glass electrode | Test solution || Reference electrode (e.g. calomel)

- The potential of the glass electrode depends only on $[H^+]$ of the solution.
- Nernst equation:

$$E = E^0 - 0.0591 \text{ pH}$$

Thus, comparing the measured EMF with a standard solution of known pH allows pH calculation.

Glass electrode



Structure of a Glass Electrode

1. Thin glass membrane (the bulb)

1. Made of special alkali-lime-silicate glass that is selectively permeable to H^+ ions.
2. The potential develops across this thin hydrated glass layer due to **ion exchange** (H^+ in the test solution vs. H^+ inside).

2. Inside the bulb

1. Contains **1 M HCl solution** (or sometimes 0.1 M).
2. A **silver wire coated with AgCl (Ag/AgCl internal electrode)** is dipped in this HCl.
3. This acts as the **internal reference electrode** with a stable potential.

Outside the bulb

The glass membrane is in contact with the **unknown solution** whose pH we want to measure.

A **reference electrode** (like SCE or external Ag/AgCl) is placed in the test solution to complete the circuit.

How it Works

At the **glass membrane**, H^+ ions from the test solution exchange with alkali metal ions (Na^+ , Li^+) in the glass surface layer.

This creates a potential difference proportional to the difference in **$[\text{H}^+]$ inside vs. outside**. Since the inside solution (1 M HCl) is constant, the potential depends only on the external solution pH.

The internal Ag/AgCl electrode provides a stable contact to measure this potential.

This potential difference is proportional to the difference in $[H^+]$ (pH) between the inner solution (fixed) and the outer solution (unknown).

Potential Expression

The potential of the glass electrode can be written as:

$$E_{\text{glass}} = E^{\circ} + \frac{0.0591}{n} \log a_{H^+}(\text{outside})$$

Since the inside solution has constant activity of H^+ , the variable part is only the **outside** $[H^+]$.

Thus,
$$E_{\text{glass}} = \text{constant} - 0.0591 \cdot \text{pH (outside)}$$

Correction for Real Behavior

- At **very low pH (<1)** → acid error (electrode saturates with H^+).
- At **very high pH (>12)** → alkaline error (Na^+ interference).
- But in the normal range, it's **accurately proportional only to $[H^+]$** .

This is the general working principle of pH meter, please read the link below

<https://microbenotes.com/ph-meter-principle-parts-procedure-types-uses-examples/>

◆ 3. General Formula for EMF and pH

If the reference electrode has constant potential E_{ref} :

$$E_{\text{cell}} = E_{\text{ref}} - 0.0591 \cdot \text{pH}$$

or rearranged:

$$\text{pH} = \frac{E_{\text{ref}} - E_{\text{cell}}}{0.0591}$$

(at 25 °C, for $n = 1$).

◆ 4. Practical Example

Suppose you use a **saturated calomel electrode (SCE)** as reference ($E_{\text{ref}} = +0.241 \text{ V}$).

You measure $E_{\text{cell}} = 0.450 \text{ V}$.

$$\text{pH} = \frac{0.241 - 0.450}{0.0591} = -3.53$$

(Negative pH means highly acidic solution).

Potentiometric Titration

What is a Potentiometric Titration?

It's a **titration method** where instead of using a color indicator, we measure the **electromotive force (EMF)** of a suitable electrochemical cell to detect the equivalence point.

- In normal titrations: endpoint = color change.
- In potentiometric titrations: endpoint = sudden change (jump) in EMF.

◆ 2. General Cell Setup

A potentiometric titration cell usually has:

- **Indicator electrode** → responds to the ion being titrated (e.g., glass electrode for H^+ , metal electrode for Ag^+ , etc.).
- **Reference electrode** → has a fixed potential (e.g., calomel, Ag/AgCl).
- **Voltmeter** → records EMF during titration.

Indicator electrode | Solution || Reference electrode

Types of Potentiometric Titrations

(a) Acid–Base Titrations

- Use **glass electrode** as indicator.
- Example: HCl titrated with NaOH.
- EMF vs. volume curve shows S-shaped plot; steep rise at equivalence.

(b) Redox Titrations

Indicator electrode = inert Pt electrode (responds to redox couple).

Example: Fe^{2+} titrated with KMnO_4 .

Sharp jump in EMF at equivalence.

(c) Precipitation Titrations

Indicator electrode = metal electrode reversible with ion (e.g., Ag electrode for Cl^-).

Example: Cl^- titrated with AgNO_3 .

Potential increases suddenly when Cl^- is completely precipitated as AgCl

(d) Complexometric Titrations

Metal ions titrated with ligands (like EDTA).

Ion-selective electrodes used.

Fig. Potentiometric titration of Sodium Chloride against Silver Nitrate Solution

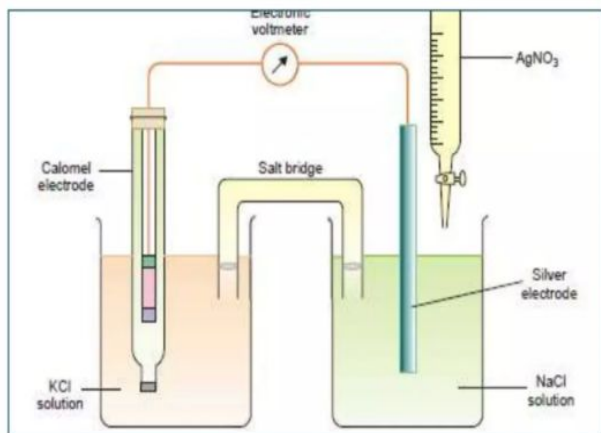
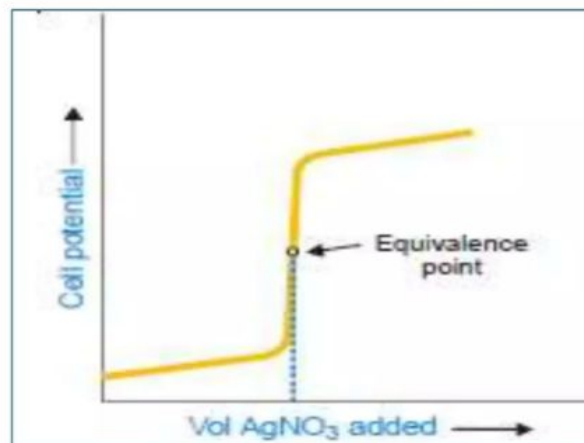


Fig. Potentiometric Titration Curve



Principle

The key reaction during titration:

- $\text{Ag}^+ + \text{Cl}^- \rightarrow \text{AgCl(s)} \downarrow$
- As AgNO₃ is added, **Cl⁻ decreases** because it precipitates as AgCl.
- Before equivalence: [Ag⁺] is very small (almost all added Ag⁺ consumed by Cl⁻).
- At equivalence: Cl⁻ is exactly consumed, so free [Ag⁺] rises sharply.
- After equivalence: [Ag⁺] in excess increases linearly with added titrant.

Electrode Potential (Nernst Equation)

For the **silver electrode** :

$$E = E_{\text{Ag}^+/\text{Ag}}^\circ + 0.0591 \log[\text{Ag}^+]$$

So, the **potential depends only on [Ag⁺]**, which in turn depends on [Cl⁻] via the solubility product.

Application of EMF measurement

1. Determination of equilibrium constant
- 2 Determination of solubility of sparingly soluble salt
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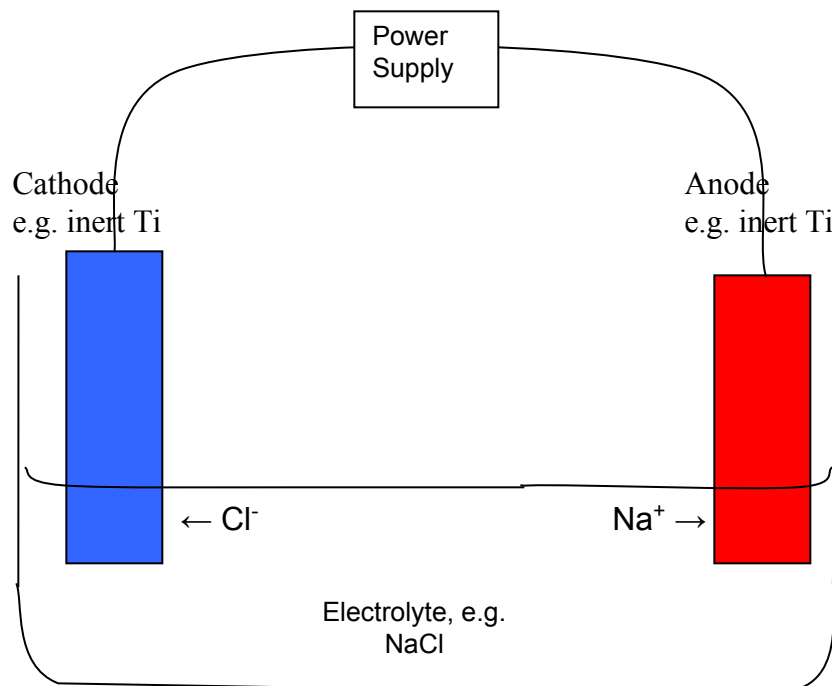
How? Detail description with picture as per required

Applications of Cells

Electrolytic cells

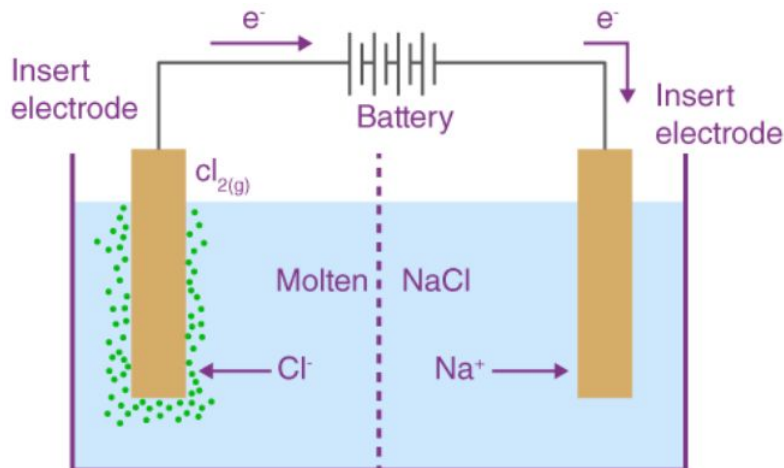
Electrolytic cells

- Use an applied voltage to carry out a nonspontaneous chemical reaction $\Rightarrow \Delta G > 0$
- Electric current supplied by an external source



- External source must provide a greater potential than that for the spontaneous reverse reaction
- **Electrolysis** = process in which electrical energy is used to cause a non-spontaneous chemical reaction to occur

Molten sodium chloride (NaCl) can be subjected to electrolysis with the help of an electrolytic cell, as illustrated below.



Here, two inert electrodes (Pt) are dipped into molten sodium chloride (which contains dissociated Na⁺ cations and Cl⁻ anions). When an electric current is passed into the circuit, the cathode becomes rich in electrons and develops a negative charge. The positively charged sodium cations are now attracted towards the negatively charged cathode. This results in the formation of metallic sodium at the cathode.

Simultaneously, the chlorine atoms are attracted to the positively charged anode. This results in the formation of chlorine gas (Cl₂) at the anode (which is accompanied by the liberation of 2 electrons, finishing the circuit). The associated chemical equations and the overall cell reaction are provided below.

Reaction at Cathode: $[\text{Na}^+ + \text{e}^- \rightarrow \text{Na}] \times 2$

Reaction at Anode: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$

Cell Reaction: $2\text{NaCl} \rightarrow 2\text{Na} + \text{Cl}_2$

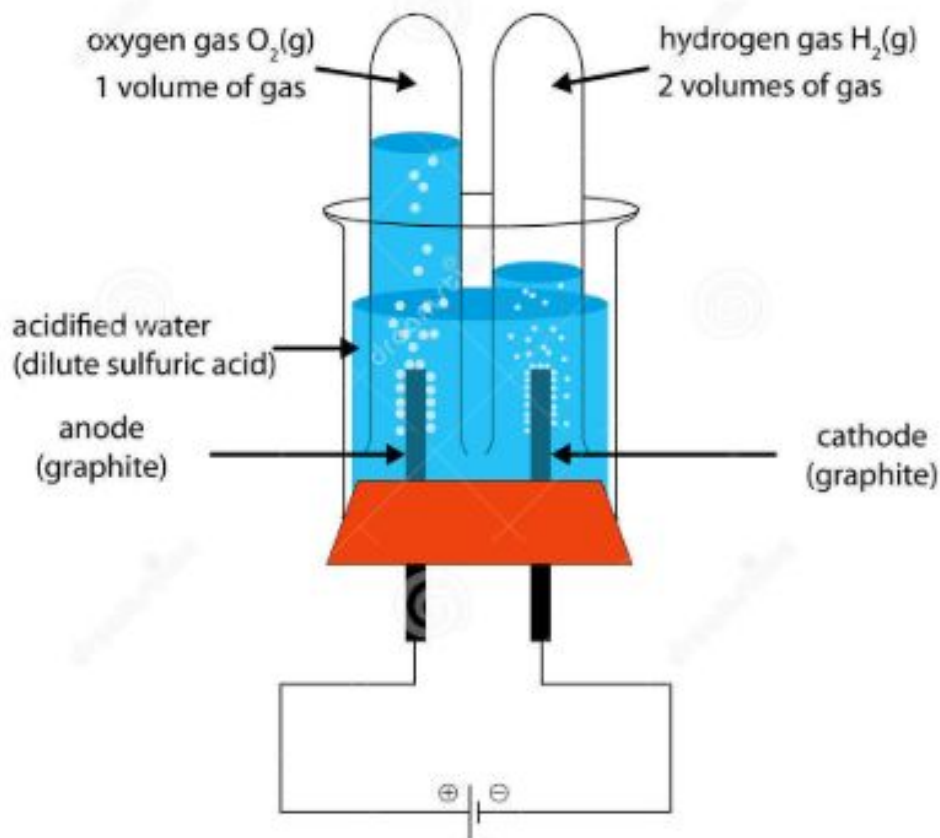
Thus, molten sodium chloride can be subjected to electrolysis in an electrolytic cell to generate metallic sodium and chlorine gas as the products.

Applications of Electrolytic Cells

❖ **Electrolytic cell:** Electrolysis, electroplating....etc.

- ✓ The primary application of electrolytic cells is for the production of oxygen gas and hydrogen gas from water.
- ✓ Another notable application of electrolytic cells is in electroplating, which is the process of forming a thin protective layer of a specific metal on the surface of another metal.
- ✓ They are also used for the extraction of aluminium from bauxite.
- ✓ The electrorefining of many non-ferrous metals is done with the help of electrolytic cells.
- ✓ Such electrochemical cells are also used in electrowinning processes.
- ✓ It can be noted that the industrial production of high-purity copper, high-purity zinc, and high-purity aluminium is almost always done through electrolytic cells.

Electrolysis of water



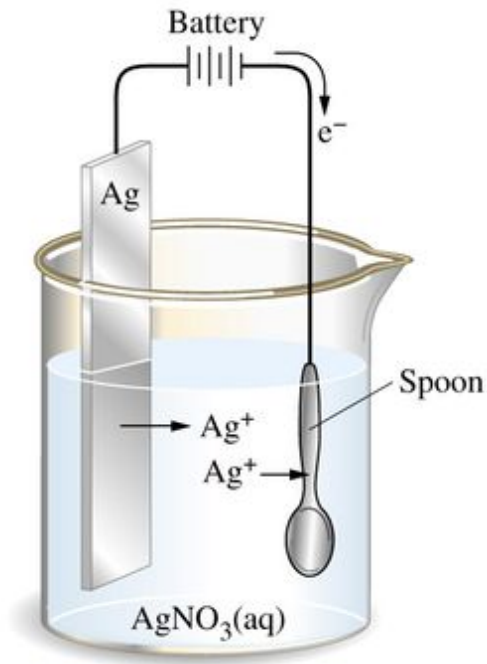
At the cathode: $4e^- + 4H^+(aq) \rightarrow 2H_2(g)$

At the anode: $2O^{2-}(aq) - 4e^- \rightarrow O_2(aq)$

Overall: $2O^{2-}(aq) + 4H^+(aq) \rightarrow 2H_2(g) + O_2(g)$

- Electrolysis of water is the decomposition of water into oxygen and hydrogen gas due to the passage of an electric current.
- The reaction has a standard potential of -1.23 V , meaning it ideally requires a potential difference of 1.23 volts to split water.
- In a non-spontaneous reaction the E_{net}^0 is always less than zero.

Electroplating



Electroplating is a procedure that uses electrolysis to apply a thin layer of a metal over the surface of another metal.